TestLink

# Test Specification

## Test Project: UniShare
## Test Suite: UniShare - Aland - Registration, Login & Admin Approval (Iteration 2)

Printed by TestLink on 26/10/2026

2026 Group UniShare

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## Scope

UniShare - university course sharing platform (Next.js / tRPC / Prisma). Group project for SECJ3343 Project 1 (Aland, Karoz, Parwar, Rawa). Source: github.com/Seganation/unishare

# 1.Test Suite : Aland - Registration, Login & Admin Approval (Iteration 2)

**Tester:** Aland (GitHub Aland95)
**Software under test:** UniShare main, authentication implementation b124314.
**Test basis:** assignment/UniShare Iteration 2.md detailed Use Cases 1-3: UC-S2-001 Student Registration, UC-S2-002 Admin Approval, UC-S2-003 Student Login; src/server/api/routers/user.ts, admin.ts, src/server/auth/config.ts, middleware.ts, src/app/(auth) and src/app/(admin), src/app/(student)/layout.tsx, src/app/api/uploadthing/core.ts, src/lib/email.ts, prisma/schema.prisma.
**Techniques:** Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table, State Transition, Use Case, Error Guessing.

**Common test data:** From prisma/seed.ts: Admin User, admin@unishare.com (ADMIN); Sarah Ahmed, sarah@student.utm.my (APPROVED); John Doe, john@student.utm.my (APPROVED); Alice Wong, alice@student.utm.my (PENDING); Michael Chen, michael@student.utm.my (APPROVED, Faculty of Engineering). All seeded passwords are admin123. Admin, Sarah, John and Alice use Universiti Teknologi Malaysia / Faculty of Computing. Seed passwords predate the new-registration uppercase rule; login does not apply registration password complexity.

**Execution setup:** UniShare runs on http://localhost:3000 with a disposable seeded Postgres database, configured UploadThing and a controlled SMTP capture/stub. Restore the initial fixture before each independent case; never reseed the group's TestLink database. The new-applicant baseline is name 'Aland QA', unused email aland.qa.01@example.com, password and confirmation 'Test1234', seeded UTM and Computing IDs obtained from user.getUniversities, and a successfully uploaded student-id.jpg image of 1 MiB. API baseline studentIdUrl is https://utfs.io/f/aland-qa-id.jpg; it is a validation fixture, not a promised live image. Use a fresh aland.qa.N@example.com email for every accepted registration within a case. API steps use a local tRPC client/server caller with the indicated session and exact JSON input; inspect responses in developer tools. Database checks use Prisma Studio or a local query. File sizes are exact bytes; 1 MiB = 1048576 bytes.

**Oracle policy:** Current code supplies URLs, limits and exact application messages. Requirements explicitly labelled document-only retain the iteration's business expectation and identify the missing check; do not mark them passed from code inspection. Browser-native validation and UploadThing errors vary by version; assert validity/error state and the application's literal 'Upload failed: ' prefix, rather than inventing library text. Numbered steps are generated by load.py. All cases remain Not Run for Project 1.

## 1.1.Test Suite : TS-A1 Registration Form and Academic Selections

UC-S2-001 form paths, cascading selections, required inputs and ID replacement. EP, Use Case, State Transition.

| **Test Case US-42: Public registration displays all mandatory controls** [Version : 1] | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Public registration displays all mandatory controls.<br><br>**Test design technique:** Use Case (registration entry), EP (anonymous visitor)<br><br>**Requirements:** REQ-A-REG-01, REQ-A-REG-10 | | |
| Preconditions:<br><br>Fresh anonymous browser; seeded universities available; no file uploaded. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open /login and click 'Sign Up Here'. | URL is /signup; heading 'Join UNIShare' is displayed. |
| 2 | Inspect all form controls. | Labels 'Full Name', 'University Email', 'Password', 'Confirm Password', 'University', 'Faculty', 'Student ID Photo' are displayed; password fields are masked. |
| 3 | Inspect 'Create Account' before uploading. | 'Create Account' is disabled; no registration request or user record is created. |
| Execution type: | Manual | |

| | |
|---|---|
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-01: Public registration form and required fields [Version : 1]<br>REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1] |
| Keywords: | None |

| **Test Case US-43: University list is alphabetical and faculties are scoped [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>University list is alphabetical and faculties are scoped.<br><br>**Test design technique:** EP (unselected/selected university), Use Case<br><br>**Requirements:** REQ-A-REG-02 | | |
| Preconditions:<br><br>Anonymous /signup; original seed contains Universiti Malaya, Universiti Sains Malaysia, Universiti Teknologi Malaysia. | | |
| #: | Step actions: | Expected Results: |
| 1 | Inspect university and faculty before selection. | Placeholders 'Select university' and 'Select faculty'; Faculty is disabled. |
| 2 | Open University and select Universiti Teknologi Malaysia; inspect user.getUniversities response. | Universities ascend by name: Malaya, Sains Malaysia, Teknologi Malaysia; UTM faculty options ascend Computing, Education, Engineering, Management, Science. |
| 3 | Select Faculty of Computing. | Selected faculty is 'Faculty of Computing'; no Malaya-specific Faculty of Computer Science and Information Technology option appears. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-REG-02: University and faculty selection [Version : 1] | |
| Keywords: | None | |

| **Test Case US-44: Changing university clears the previously selected faculty [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Changing university clears the previously selected faculty.<br><br>**Test design technique:** State Transition (UTM selected -> Malaya selected), Error Guessing<br><br>**Requirements:** REQ-A-REG-02, REQ-A-REG-11 | | |
| Preconditions:<br><br>Anonymous /signup, baseline valid text fields and uploaded ID; UTM / Faculty of Computing selected. | | |
| #: | Step actions: | Expected Results: |
| 1 | Change University to Universiti Malaya. | Faculty resets to 'Select faculty'; UTM Faculty of Computing is no longer an option. |
| 2 | Click 'Create Account' without selecting a new faculty. | Native required-field validation blocks submission at 'Faculty'; no user.register request is sent. |

| | | |
|---|---|---|
| 3 | Select Faculty of Computer Science and Information Technology. | New faculty belongs to Malaya; required-field validity is restored. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-REG-02: University and faculty selection [Version : 1]<br>REQ-A-REG-11: Required academic selections [Version : 1] | |
| Keywords: | None | |

| **Test Case US-45: Each empty required registration input blocks submission [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Each empty required registration input blocks submission.<br><br>**Test design technique:** Decision Table (one missing mandatory input per row), EP negative<br><br>**Requirements:** REQ-A-REG-01, REQ-A-REG-11 | | |
| Preconditions:<br><br>Anonymous /signup with valid baseline and uploaded ID; restore baseline before each row; inspect Network panel. | | |
| #: | Step actions: | Expected Results: |
| 1 | For each row separately, clear Full Name, University Email, Password, or Confirm Password and click 'Create Account'. | Each empty required field fails native validity; no user.register request and no applicant is created; exact browser bubble wording is browser-dependent. |
| 2 | Restore baseline; set University back to 'Select university' and submit. | University is invalid, Faculty is cleared; no registration request. |
| 3 | Restore baseline; set Faculty to 'Select faculty' and submit. | Faculty is invalid; no registration request; 'Create Account' remains available after correction. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-REG-01: Public registration form and required fields [Version : 1]<br>REQ-A-REG-11: Required academic selections [Version : 1] | |
| Keywords: | None | |

| **Test Case US-46: Remove uploaded student ID then replace it [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Remove uploaded student ID then replace it.<br><br>**Test design technique:** State Transition (uploaded -> absent -> uploaded)<br><br>**Requirements:** REQ-A-REG-10, REQ-A-REG-15 | | |
| Preconditions:<br><br>Anonymous /signup with baseline fields and a successful 1 MiB student-id.jpg upload; keep returned first URL. | | |
| #: | Step actions: | Expected Results: |
| 1 | Check completed upload and click 'Remove'. | 'Student ID uploaded' disappears; uploader returns; 'Create Account' is disabled. |

| | | |
|---|---|---|
| 2 | Upload replacement-id.png (valid 1 MiB PNG). | 'Student ID uploaded' and 'Remove' reappear; 'Create Account' is enabled. |
| 3 | Submit and inspect user.register payload and stored row. | Payload/studentIdUrl use replacement URL, not old URL; UI displays 'Registration Successful!'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1]<br>REQ-A-REG-15: Upload removal and recovery [Version : 1] | |
| Keywords: | None | |

## 1.2.Test Suite : TS-A2 Registration Input Validation

UC-S2-001 EP, BVA, Decision Table and tampering. Business-rule gaps remain explicit expected failures.

| **Test Case US-47: Name minimum boundaries at 1, 2 and 3 characters [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Name minimum boundaries at 1, 2 and 3 characters.<br><br>**Test design technique:** BVA (min-1/min/min+1 = 1/2/3)<br><br>**Requirements:** REQ-A-REG-03 | | |
| Preconditions:<br><br>Public user.register client; baseline valid payload; unused aland.qa.06a/b/c@example.com per row. | | |
| #: | Step actions: | Expected Results: |
| 1 | Call user.register with name 'A' (1 character). | Validation issue 'Name must be at least 2 characters'; no user is inserted. |
| 2 | Repeat with name 'Al' (2), fresh email. | success=true; message 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; name stored as Al. |
| 3 | Repeat with name 'Ali' (3), fresh email. | success=true with the same exact success message; name stored as Ali. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-REG-03: Minimum name length [Version : 1] | |
| Keywords: | None | |

| **Test Case US-48: Document maximum name boundary at 99, 100 and 101 characters [Version : 1]** | |
|---|---|
| Author: | aland |
| Summary:<br><br>Document maximum name boundary at 99, 100 and 101 characters.<br><br>**Test design technique:** BVA (max-1/max/max+1 = 99/100/101), Error Guessing<br><br>**Requirements:** REQ-A-REG-04 | |
| Preconditions: | |

Public user.register with valid baseline; fresh aland.qa.07a/b/c@example.com per input. Document-derived case; current code lacks max(100).

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Register with name exactly 'A' repeated 99 times and a fresh email. | Accepted: success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. |
| 2 | Register with name exactly 'A' repeated 100 times and a fresh email. | Accepted: same success message; full 100-character name stored. |
| 3 | Register with name exactly 'A' repeated 101 times and a fresh email. | Document expectation: validation rejection, no new user. Current implementation instead accepts it; record discrepancy against the 100-character rule. No app maximum-name error text exists. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-A-REG-04: Maximum name length of 100 (document-only gap) [Version : 1] |
| Keywords: | None |

**Test Case US-49: Email invalid partitions and valid nonuniversity domain [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Email invalid partitions and valid nonuniversity domain.

**Test design technique:** EP (empty/missing-at/missing-domain/valid)

**Requirements:** REQ-A-REG-05

Preconditions:

Public user.register; baseline except email; aland.qa.08@example.com unused. Anonymous /signup available for UI check.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Individually submit email '', 'aland.example.com', and 'aland@' through API. | Every input returns validation issue 'Invalid email address'; no applicants created. |
| 2 | Set UI University Email to 'aland.example.com' with other fields valid; submit. | Native email validity fails; user.register is not called; browser wording is not app-controlled. |
| 3 | Register through API using aland.qa.08@example.com. | Accepted: success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; no university-domain restriction is applied. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-05: Email syntax [Version : 1] |
| Keywords: | None |

**Test Case US-50: Existing approved and pending email cannot register twice [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Existing approved and pending email cannot register twice.

**Test design technique:** EP (duplicate APPROVED/PENDING), Error Guessing

**Requirements:** REQ-A-REG-06

Preconditions:

Original seed: Sarah APPROVED and Alice PENDING; baseline password Test1234 meets registration rules; capture their initial rows.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call user.register with sarah@student.utm.my and otherwise valid baseline. | Error 'User with this email already exists'; Sarah's role, password hash and details unchanged. |
| 2 | Repeat with alice@student.utm.my. | Same error 'User with this email already exists'; one Alice row remains PENDING with original details. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-06: Unique registration email and error recovery [Version : 1] |
| Keywords: | None |

**Test Case US-51: Password length boundaries at 7, 8 and 9 characters [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Password length boundaries at 7, 8 and 9 characters.

**Test design technique:** BVA (min-1/min/min+1 = 7/8/9)

**Requirements:** REQ-A-REG-07

Preconditions:

Anonymous /signup with baseline including uploaded ID; reset form between accepted rows, fresh aland.qa.10a/b@example.com.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Set password and confirmation to 'Test123' (7), click 'Create Account'. | 'Password must be at least 8 characters'; no registration request; form reenabled. |
| 2 | Set both to 'Test1234' (8), submit with fresh email. | 'Registration Successful!'; applicant created PENDING. |
| 3 | Return to /signup; fill baseline with fresh email and both passwords 'Test12345' (9); submit. | 'Registration Successful!'; 9-character password is accepted. |
| 4 | Call API user.register with 'Test123' (7) in otherwise valid baseline. | Server independently rejects with 'Password must be at least 8 characters'; no user inserted. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-07: Password minimum length [Version : 1] |
| Keywords: | None |

**Test Case US-52: Password complexity decision table for all three classes [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Password complexity decision table for all three classes.

**Test design technique:** Decision Table (uppercase/lowercase/digit), EP

**Requirements:** REQ-A-REG-08

Preconditions:

Public user.register; baseline valid fields; distinct fresh email for each accepted row.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Submit 'test1234' (no uppercase), 'TEST1234' (no lowercase), 'Testabcd' (no digit), each separately. | Each rejected with 'Password must contain at least one uppercase letter, one lowercase letter, and one number'; no applicant created. |
| 2 | Submit '!!!!!!!!' (none of the three required classes). | Same exact complexity error; no applicant created. |
| 3 | Submit 'Test1234' (all classes, no symbol) then 'Test123!' (all classes plus symbol), with fresh emails. | Both accepted with success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-08: Password character classes [Version : 1] |
| Keywords: | None |

**Test Case US-53: Mismatched confirmation blocks then corrected value succeeds [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Mismatched confirmation blocks then corrected value succeeds.

**Test design technique:** EP (unequal/equal), State Transition (error -> corrected)

**Requirements:** REQ-A-REG-09

Preconditions:

Anonymous /signup with baseline and uploaded ID, unused aland.qa.12@example.com; Network panel open.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Enter password 'Test1234', confirmation 'Test1235'; click 'Create Account'. | Exact error 'Passwords do not match'; no user.register request; fields become editable. |
| 2 | Change confirmation to 'Test1234' and submit again. | Error cleared; one request succeeds; 'Registration Successful!' displayed; exactly one applicant created. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-09: Password confirmation [Version : 1] |
| Keywords: | None |

**Test Case US-54: API rejects empty academic selections and invalid ID URLs [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

API rejects empty academic selections and invalid ID URLs.

**Test design technique:** EP (missing academic value / malformed URL / valid URL), Error Guessing

**Requirements:** REQ-A-REG-10, REQ-A-REG-11

Preconditions:

Public user.register; valid baseline except the field under test. Restore baseline per row; no applicant for rejected input.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Submit universityId='' then restore and submit facultyId=''. | Respectively 'University is required' and 'Faculty is required'; no user created. |
| 2 | Submit studentIdUrl='' and then 'not-a-url', separately. | Both return 'Invalid student ID URL'; no user created. |
| 3 | Submit studentIdUrl='https://example.com/id.png', all other fields valid, fresh email. | API accepts syntactically valid URL, success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. Current schema has no UploadThing-host allowlist. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-10: Required uploaded ID and URL validation [Version : 1]<br>REQ-A-REG-11: Required academic selections [Version : 1] |
| Keywords: | None |

**Test Case US-55: Tampered faculty from a different university must be rejected [Version : 1]**

Author: aland

Summary:

Tampered faculty from a different university must be rejected.

**Test design technique:** Decision Table (selected university matches faculty owner / does not match), Error Guessing

**Requirements:** REQ-A-REG-12

Preconditions:

Fetch real IDs with user.getUniversities. U=UTM, F=Universiti Malaya Faculty of Computer Science and Information Technology; baseline valid, unused aland.qa.14@example.com. Document-derived negative case.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call user.register with universityId=U and facultyId=F (both exist, but F belongs to Malaya). | Document expectation: reject with 'Invalid faculty selection' and create no applicant. Current code has no relationship check and is expected to accept this pair; record the discrepancy, not a passing result. |
| 2 | Repeat with UTM's actual Computing faculty ID and a different unused email. | Valid pair accepted with 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-12: Faculty must belong to selected university (document-only gap) [Version : 1] |
| Keywords: | None |

## 1.3.Test Suite : TS-A3 Student ID Upload Validation

UC-S2-001 public image upload: type, exact file-size boundary, file count and failure recovery. EP, BVA, Error Guessing.

| **Test Case US-56: Public image upload accepts JPG and PNG but rejects PDF [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Public image upload accepts JPG and PNG but rejects PDF.<br><br>**Test design technique:** EP (supported image / unsupported document), Use Case<br><br>**Requirements:** REQ-A-REG-13 | | |
| Preconditions:<br><br>Anonymous /signup; working UploadThing; genuine valid student-id.jpg, student-id.png and student-id.pdf, each 1 MiB; no session. | | |
| #: | Step actions: | Expected Results: |
| 1 | Upload student-id.jpg through studentIdUploader; inspect completion response. | Upload succeeds without login; response uploadedBy='registration' and fileUrl is returned; 'Student ID uploaded' displayed. |
| 2 | Click 'Remove'; upload student-id.png. | PNG accepted; 'Student ID uploaded' displayed; one active URL selected. |
| 3 | Click 'Remove'; attempt student-id.pdf. | File selection or uploader validation rejects PDF; 'Student ID uploaded' is absent and 'Create Account' disabled. Any application callback error begins 'Upload failed: '; library suffix is version-dependent. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-REG-13: Public upload accepts one image [Version : 1] | |
| Keywords: | None | |

| **Test Case US-57: ID image size boundaries at 4 MiB minus one, exact and plus one byte [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>ID image size boundaries at 4 MiB minus one, exact and plus one byte.<br><br>**Test design technique:** BVA (4194303 / 4194304 / 4194305 bytes)<br><br>**Requirements:** REQ-A-REG-14 | | |
| Preconditions:<br><br>Anonymous /signup; valid decodable PNG fixtures id-below.png=4194303 bytes, id-exact.png=4194304, id-above.png=4194305; inspect sizes on disk; working UploadThing. | | |
| #: | Step actions: | Expected Results: |
| 1 | Upload id-below.png (4194303 bytes). | Accepted; 'Student ID uploaded'; 'Create Account' enabled when upload completes. |
| 2 | Remove; upload id-exact.png (4194304 bytes). | Accepted at inclusive maximum; 'Student ID uploaded'. |
| 3 | Remove; attempt id-above.png (4194305 bytes). | Rejected over maximum; no selected URL; 'Create Account' disabled. If callback runs, application error prefix is 'Upload failed: '; do not substitute the iteration's unimplemented custom message. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |

| | |
|---|---|
| Requirements | REQ-A-REG-14: Student ID file size maximum [Version : 1] |
| Keywords: | None |

| **Test Case US-58: Two-file batch and failed upload do not enable registration [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Two-file batch and failed upload do not enable registration.<br><br>**Test design technique:** BVA (file count 1/2), Error Guessing (network failure), State Transition<br><br>**Requirements:** REQ-A-REG-13, REQ-A-REG-15 | | |
| Preconditions:<br><br>Anonymous /signup with baseline valid fields; two genuine 1 MiB images; developer-tools request interception or upload test stub available. | | |
| #: | Step actions: | Expected Results: |
| 1 | Submit a single upload batch containing both images through studentIdUploader (use API if picker limits selection). | Batch rejected by maxFileCount=1; no successful completed two-image upload. 'Create Account' remains disabled until one upload succeeds. |
| 2 | Attempt one image and inject uploader onUploadError with Error('QA upload unavailable'). | Exact UI error 'Upload failed: QA upload unavailable'; no 'Student ID uploaded'; 'Create Account' disabled. |
| 3 | Remove fault injection and retry one image. | Upload completes; 'Student ID uploaded' and 'Remove' displayed; 'Create Account' enabled. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-REG-13: Public upload accepts one image [Version : 1]<br>REQ-A-REG-15: Upload removal and recovery [Version : 1] | |
| Keywords: | None | |

## 1.4.Test Suite : TS-A4 Registration Persistence and Email

UC-S2-001 end-to-end submission, bcrypt, PENDING state, email delivery failure and form recovery. Use Case, State Transition, Error Guessing.

| **Test Case US-59: Successful applicant is PENDING with hashed password and welcome email [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Successful applicant is PENDING with hashed password and welcome email.<br><br>**Test design technique:** Use Case (registration main flow), State Transition (absent -> PENDING)<br><br>**Requirements:** REQ-A-REG-16, REQ-A-REG-17 | | |
| Preconditions:<br><br>Anonymous /signup; baseline with unused aland.qa.18@example.com and real uploaded image; controlled SMTP capture succeeds; disposable database inspection available. | | |
| #: | Step actions: | Expected Results: |
| 1 | Complete baseline fields; click 'Create Account'; capture mutation request/response. | During submission fields disable and button says 'Creating...'; response success=true with userId and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'. |

| | | |
|---|---|---|
| 2 | Observe immediate success panel and wait at least 3 seconds. | 'Registration Successful!', 'Your account has been created and is pending admin approval.', 'Redirecting to login...' displayed, then /login; no authenticated applicant session is automatically created. |
| 3 | Inspect the new User and verify bcrypt.compare('Test1234', stored password) in a local verification tool. | Exactly one user: name Aland QA, role PENDING, approvedAt null, selected UTM/Computing IDs and returned ID URL; timestamps set; password differs from plaintext, bcrypt cost 12, comparison true. |
| 4 | Inspect captured welcome mail addressed to aland.qa.18@example.com. | Subject 'Welcome to UNIShare - Registration Received'; body contains 'Your account status:' and 'PENDING'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1]<br>REQ-A-REG-17: Welcome email is best effort [Version : 1] |
| Keywords: | None |

**Test Case US-60: SMTP welcome failure preserves a successful registration [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

SMTP welcome failure preserves a successful registration.

**Test design technique:** Error Guessing (SMTP failure), State Transition

**Requirements:** REQ-A-REG-16, REQ-A-REG-17, REQ-A-REG-06

Preconditions:

Public user.register with baseline aland.qa.19@example.com unused; controlled mail transport throws Error('QA SMTP down'); restore transport afterward.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Register baseline applicant while SMTP fails. | success=true and 'Registration successful! Please wait for admin approval. You'll receive an email once approved.'; server logs 'Failed to send welcome email:'. |
| 2 | Inspect database, then retry same registration email. | One PENDING applicant with null approvedAt remains; retry returns 'User with this email already exists' and does not insert a second user. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-06: Unique registration email and error recovery [Version : 1]<br>REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1]<br>REQ-A-REG-17: Welcome email is best effort [Version : 1] |
| Keywords: | None |

**Test Case US-61: Duplicate-email server error can be corrected without reuploading [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Duplicate-email server error can be corrected without reuploading.

**Test design technique:** State Transition (submitting -> error -> success), Error Guessing

**Requirements:** REQ-A-REG-06, REQ-A-REG-16

| Preconditions: | | |
|---|---|---|
| Anonymous /signup; valid baseline and uploaded ID; email initially sarah@student.utm.my; aland.qa.20@example.com unused. | | |
| #: | Step actions: | Expected Results: |
| 1 | Submit existing Sarah email with otherwise valid Test1234 baseline. | Exact error 'User with this email already exists'; loading clears, 'Create Account' and fields reenabled; selected 'Student ID uploaded' remains. |
| 2 | Change only email to aland.qa.20@example.com and submit. | Old error clears; 'Registration Successful!'; one new PENDING applicant, original Sarah unchanged. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-REG-06: Unique registration email and error recovery [Version : 1]<br>REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1] | |
| Keywords: | None | |

## 1.5.Test Suite : TS-A5 Credential Login, Role Routing and Sessions

UC-S2-003 and Admin Login: role decision table, bad credentials, session persistence, fresh role and sign out. EP, Decision Table, State Transition, Use Case.

| **Test Case US-62: Approved student logs in with seeded credentials** [Version : 1] | | |
|---|---|---|
| Author: | aland | |
| Summary: | | |
| Approved student logs in with seeded credentials.<br>**Test design technique:** Use Case (login main flow), Decision Table (APPROVED + correct password)<br>**Requirements:** REQ-A-LOG-01 | | |
| Preconditions: | | |
| Anonymous fresh browser; seed Sarah APPROVED; password admin123; UniShare and database available. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open /login; enter sarah@student.utm.my and admin123; click 'Sign In'. | 'Signing In...' then successful overlay 'Welcome Back!' and 'Taking you to your dashboard...'. |
| 2 | Wait for redirect and inspect current session. | Final URL /courses; 'Courses' navigation visible; session user is Sarah with role APPROVED. Seeded lowercase admin123 is accepted despite new-registration uppercase rule. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-LOG-01: Valid credentials route by current role [Version : 1] | |
| Keywords: | None | |

| **Test Case US-63: Pending login cannot access student features and can sign out** [Version : 1] | |
|---|---|
| Author: | aland |
| Summary: | |
| Pending login cannot access student features and can sign out.<br>**Test design technique:** Decision Table (PENDING + correct password), State Transition (authenticated -> signed out) | |

**Requirements:** REQ-A-LOG-01, REQ-A-LOG-06

Preconditions:

Fresh browser; original seed Alice PENDING with null approvedAt; password admin123.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Sign in on /login with alice@student.utm.my / admin123. | Login succeeds, final URL /waiting-approval, heading 'Account Pending Approval'; text 'Your registration is currently under review by our admin team.'. |
| 2 | Individually open /courses, /timetable, /my-articles and /settings. | Each finishes at /waiting-approval; 'Account Pending Approval' displayed instead of student content. |
| 3 | Click 'Sign Out'; query user.getCurrentSession and revisit /courses. | Returns /login; getCurrentSession is null; /courses redirects /login with 'Sign In' button. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-LOG-01: Valid credentials route by current role [Version : 1]<br>REQ-A-LOG-06: Pending access restrictions and sign out [Version : 1] |
| Keywords: | None |

**Test Case US-64: Admin login reaches approval dashboard [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Admin login reaches approval dashboard.

**Test design technique:** Decision Table (ADMIN + correct password), Use Case (Admin Login)

**Requirements:** REQ-A-LOG-01, REQ-A-ADM-01

Preconditions:

Anonymous browser; Admin User admin@unishare.com / admin123 has ADMIN role.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Sign in with admin@unishare.com / admin123. | 'Welcome Back!' then redirects through /courses to /admin/approvals; 'UNIShare Admin' and 'Student Approvals' visible. |
| 2 | Open /admin and then /courses directly. | Both finish at /admin/approvals; 'Student Approvals' visible; admin is not left on the student page. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-01: Admin route and API authorization [Version : 1]<br>REQ-A-LOG-01: Valid credentials route by current role [Version : 1] |
| Keywords: | None |

**Test Case US-65: Unknown email and wrong password have the same safe error [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Unknown email and wrong password have the same safe error.

**Test design technique:** EP (unknown account / bad password), Decision Table negative

**Requirements:** REQ-A-LOG-02

Preconditions:

Anonymous /login; missing.qa@example.com absent; Sarah exists; query user.getCurrentSession after each failure.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Submit missing.qa@example.com / admin123. | UI 'Invalid email or password'; remains /login; 'Sign In' and fields reenabled; getCurrentSession returns null. |
| 2 | Submit sarah@student.utm.my / Wrong123!. | Identical 'Invalid email or password'; no account-existence disclosure; no session. |
| 3 | Correct password to admin123 and submit. | 'Welcome Back!' then /courses; previous error cleared. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-LOG-02: Generic invalid-credentials errors [Version : 1] |
| Keywords: | None |

**Test Case US-66: Missing and malformed login fields cannot authenticate [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Missing and malformed login fields cannot authenticate.

**Test design technique:** EP (empty email / empty password / malformed email), Error Guessing

**Requirements:** REQ-A-LOG-03

Preconditions:

Anonymous /login, Network panel open; local authorize test harness available for bypassing native validation.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Submit with Email Address empty/password admin123; then valid Sarah email/empty Password. | Each missing field fails required validity; no signIn request; 'Email Address', 'Password' and 'Sign In' remain displayed. |
| 2 | Submit email 'sarah-at-student.utm.my', password admin123. | Native type=email validity fails; no signIn request; exact browser validation wording is browser-dependent. |
| 3 | Invoke authorize with email='' and password='admin123', then with valid Sarah email and password='', using local harness. | Each throws 'Missing email or password'; no user is authorized. External login UI maps auth errors to 'Invalid email or password'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-LOG-03: Required login fields [Version : 1] |
| Keywords: | None |

**Test Case US-67: Existing sessions bypass login and signup by role [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Existing sessions bypass login and signup by role.

**Test design technique:** Decision Table (APPROVED/PENDING/ADMIN x /login,/signup)

**Requirements:** REQ-A-LOG-04, REQ-A-LOG-01

Preconditions:

Three separate browser profiles already signed in as Sarah APPROVED, Alice PENDING and Admin ADMIN with seeded credentials.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | In Sarah's profile navigate to /login and /signup separately. | Both end at /courses; 'Courses' visible; no new login/registration form remains. |
| 2 | In Alice's profile navigate to /login and /signup separately. | Both end at /waiting-approval with 'Account Pending Approval'. |
| 3 | In Admin's profile navigate to /login and /signup separately. | Both finish at /admin/approvals with 'Student Approvals'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-A-LOG-01: Valid credentials route by current role [Version : 1]<br>REQ-A-LOG-04: Session persistence and authenticated-page redirects [Version : 1] |
| Keywords: | None |

**Test Case US-68: JWT session survives reload and another tab without exposing password [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

JWT session survives reload and another tab without exposing password.

**Test design technique:** Use Case (session continuation), Error Guessing (password disclosure)

**Requirements:** REQ-A-LOG-04

Preconditions:

Sarah logged in at /courses in one normal browser profile; developer tools and user.getCurrentSession client available.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Reload /courses, then open /courses in a new tab of the same profile. | Both render authenticated 'Courses'; neither asks for 'Sign In'. |
| 2 | Inspect /api/auth/session and user.getCurrentSession. | user.id matches Sarah, role APPROVED, universityId and facultyId match seed; no password/plaintext/hash field is exposed. |
| 3 | Inspect authentication session configuration and persisted session cookie existence. | JWT session cookie exists; auth/config.ts specifies strategy 'jwt'. No new database Session row or Remember me control is required by the implemented flow. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-LOG-04: Session persistence and authenticated-page redirects [Version : 1] |
| Keywords: | None |

**Test Case US-69: Pending session observes admin approval without signing in again [Version : 1]**

| Author: | aland | |
|---|---|---|
| Summary:<br><br>Pending session observes admin approval without signing in again.<br><br>**Test design technique:** State Transition (PENDING -> APPROVED), Use Case (check approval status)<br><br>**Requirements:** REQ-A-LOG-05, REQ-A-ADM-06 | | |
| Preconditions:<br><br>Alice PENDING signed in at /waiting-approval in profile A; Admin signed in profile B; restore Alice afterward. | | |
| #: | Step actions: | Expected Results: |
| 1 | In profile B click Alice's 'Approve' and confirm 'Are you sure you want to approve this user?'. | Alice removed from 'Pending Approvals'; database role APPROVED and approvedAt set. |
| 2 | In profile A, without sign out or password entry, navigate to /courses and fetch fresh /api/auth/session. | 'Courses' is accessible; fresh session role APPROVED for same Alice id, reflecting current database role. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1]<br>REQ-A-LOG-05: Fresh approval status without a new login [Version : 1] | |
| Keywords: | None | |

## 1.6.Test Suite : TS-A6 Admin Access and Applicant Review

UC-S2-002 authorization matrix, list filtering/order, student ID inspection, statistics and document-only search/sort. Decision Table, EP, BVA, Use Case.

| **Test Case US-70: Anonymous and nonadmin users cannot access admin pages or procedures [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Anonymous and nonadmin users cannot access admin pages or procedures.<br><br>**Test design technique:** Decision Table (anonymous/APPROVED/PENDING x admin read/approve/reject)<br><br>**Requirements:** REQ-A-ADM-01 | | |
| Preconditions:<br><br>Original seed; use anonymous, Sarah APPROVED and Alice PENDING profiles. Capture Alice's id, role and approvedAt. Mutations target Alice; run without ADMIN session. | | |
| #: | Step actions: | Expected Results: |
| 1 | As anonymous open /admin/approvals; call admin.getPendingUsers, getAllUsers, getDashboardStats, approveUser({userId:AliceId}) and rejectUser({userId:AliceId}). | Page ends at /login showing 'Sign In'; each API request returns UNAUTHORIZED; Alice remains PENDING. |
| 2 | As Sarah repeat the page and the five API calls. | Page ends at /courses with 'Courses'; each API returns FORBIDDEN 'Only administrators can perform this action'; no user data or mutation allowed. |
| 3 | As Alice repeat the page and the five API calls. | Page ends at /waiting-approval with 'Account Pending Approval'; each API returns FORBIDDEN 'Only administrators can perform this action'; Alice remains PENDING, approvedAt null. |
| Execution type: | Manual | |

| | |
|---|---|
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-01: Admin route and API authorization [Version : 1] |
| Keywords: | None |

| **Test Case US-71: Pending list excludes approved/admin accounts, orders newest first and handles zero [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary: | | |
| Pending list excludes approved/admin accounts, orders newest first and handles zero.<br><br>**Test design technique:** EP (PENDING/APPROVED/ADMIN), BVA (list size 0/1/2)<br><br>**Requirements:** REQ-A-ADM-02 | | |
| Preconditions: | | |
| Admin signed in; disposable fixture with Alice as sole PENDING, other seeded users unchanged. Can add new PENDING applicant 'Zara QA', zara.qa@example.com, via baseline register with later createdAt; controlled SMTP. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open /admin/approvals and inspect admin.getPendingUsers response with Alice sole PENDING. | One Alice card under 'Pending Approvals'; Sarah, John and Admin excluded; response has no password field. |
| 2 | Register Zara QA after Alice; refresh admin page and inspect admin.getAllUsers response. | Exactly two pending cards: Zara then Alice (createdAt descending); getAllUsers includes roles but no password field for any user. |
| 3 | Approve both disposable pending users as Admin and wait for refreshed queries. | Zero cards; 'All caught up!' and 'There are no pending approvals at the moment.' displayed. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-02: Pending list and safe user data [Version : 1] | |
| Keywords: | None | |

| **Test Case US-72: Admin sees applicant identity and opens the stored student ID [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary: | | |
| Admin sees applicant identity and opens the stored student ID.<br><br>**Test design technique:** Use Case (review student identity), EP (pending applicant)<br><br>**Requirements:** REQ-A-ADM-03 | | |
| Preconditions: | | |
| Admin at /admin/approvals; Alice PENDING, original seed URL https://utfs.io/f/example-alice-id.jpg. Seed URL may not host a real image; use a known reachable controlled image URL in Alice's local fixture for rendering if needed and record it. | | |
| #: | Step actions: | Expected Results: |
| 1 | Inspect Alice's card. | 'Alice Wong', 'alice@student.utm.my', 'University:', 'Universiti Teknologi Malaysia', 'Faculty:', 'Faculty of Computing', 'Registered:', 'Student ID', 'Approve', 'Reject' are displayed; date matches local rendering of createdAt. |

| | | |
|---|---|---|
| 2 | Click 'View full size →'; inspect new-tab URL. | Opens exact stored studentIdUrl in a new tab; the app does not alter or substitute it. Seed-link availability is a fixture limitation, not evidence of rendering success. |
| 3 | Close new tab; click the Student ID image/preview. | Again opens the same stored studentIdUrl; admin page remains at /admin/approvals. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-03: Review identity details and full-size student ID [Version : 1] | |
| Keywords: | None | |

| **Test Case US-73: Dashboard role counts refresh after approve and reject [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Dashboard role counts refresh after approve and reject.<br><br>**Test design technique:** State Transition (role counts after decisions), Use Case<br><br>**Requirements:** REQ-A-ADM-04 | | |
| Preconditions:<br><br>Admin signed in; eight original seeded users (1 ADMIN, 6 APPROVED, 1 PENDING) plus newly registered PENDING 'Aland QA', aland.qa.32@example.com. Initial counts: total9, pending2, approved6, admin1; no other mutations. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open /admin/approvals; compare four cards to admin.getDashboardStats. | 'Pending Approvals'=2, 'Approved Users'=6, 'Total Users'=9, 'Administrators'=1. |
| 2 | Approve Alice and confirm; wait for refetch without manual reload. | Counts pending1, approved7, total9, administrators1; Alice card removed. |
| 3 | Reject Aland QA, leave reason blank, confirm; wait for refetch. | Counts pending0, approved7, total8, administrators1; 'All caught up!' displayed. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-ADM-04: Dashboard totals refresh after decisions [Version : 1] | |
| Keywords: | None | |

| **Test Case US-74: Documented pending-applicant search and sorting are available [Version : 1]** | |
|---|---|
| Author: | aland |
| Summary:<br><br>Documented pending-applicant search and sorting are available.<br><br>**Test design technique:** Use Case (search/filter alternative), EP (name/email/no match)<br><br>**Requirements:** REQ-A-ADM-05 | |
| Preconditions:<br><br>Admin signed in; Alice and newly registered PENDING Zara QA (zara.qa@example.com) exist with distinct dates and universities. Document-only case: current UI has no search/sort controls. | |

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open /admin/approvals and locate name/email filtering and date/name/university sorting described in Iteration 2 Use Case 2 step 8. | Document expectation: controls exist. Current page only displays 'Pending Approvals' cards without these controls; record implementation gap if absent; no exact labels are specified by code. |
| 2 | If controls exist, filter 'Alice', 'zara.qa@example.com' and 'no-such-applicant', clearing between inputs. | Document expectation: Alice only, Zara only, then zero applicants; reset restores both. Current absence prevents these actions and must not be reported as passed. |
| 3 | If available, sort by date, name and university in each supported direction. | Document expectation: stable list ordered by selected field, containing only PENDING applicants. Current code offers only fixed createdAt descending order. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-ADM-05: Search/filter and sorting (document-only gap) [Version : 1] | |
| Keywords: | None | |

## 1.7.Test Suite : TS-A7 Approve Student Registration

UC-S2-002 approval: cancel/confirm, role/timestamp transition, email, invalid targets and sequential replay. State Transition, Decision Table, Error Guessing.

| **Test Case US-75: Cancel approval leaves applicant pending [Version : 1]** | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Cancel approval leaves applicant pending.<br><br>**Test design technique:** Decision Table (approval confirmation = cancel), State Transition (no transition)<br><br>**Requirements:** REQ-A-ADM-06 | | |
| Preconditions:<br><br>Admin signed in at /admin/approvals; Alice PENDING, approvedAt null; initial counts and mail capture noted; Network panel open. | | |
| #: | Step actions: | Expected Results: |
| 1 | Click Alice's 'Approve'. | Browser confirmation reads 'Are you sure you want to approve this user?'. |
| 2 | Choose Cancel and reload. | No admin.approveUser mutation; Alice card still present, role PENDING, approvedAt null; counts and captured mail unchanged. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1] | |
| Keywords: | None | |

| **Test Case US-76: Confirm approval updates role, timestamp and email [Version : 1]** | |
|---|---|
| Author: | aland |
| Summary:<br><br>Confirm approval updates role, timestamp and email.<br><br>**Test design technique:** State Transition (PENDING -> APPROVED), Use Case main flow | |

**Requirements:** REQ-A-ADM-06, REQ-A-ADM-08

Preconditions:

Admin at /admin/approvals; Alice PENDING, approvedAt null; successful controlled SMTP transport can delay response 2 seconds; record start time and NEXT_PUBLIC_APP_URL=http://localhost:3000.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Click Alice's 'Approve' and confirm 'Are you sure you want to approve this user?'. | While mutation pending, selected button says 'Approving...' and both Approve/Reject buttons disable; no second UI decision can be submitted. |
| 2 | Inspect response, database and refreshed list. | success=true, 'Alice Wong has been approved successfully'; role APPROVED, approvedAt between start and completion times; Alice card removed. No success toast is required by current UI. |
| 3 | Inspect captured approval email to alice@student.utm.my. | Exact subject starts with U+1F389 (party popper) and a space, followed by 'Your UNIShare Account Has Been Approved!'; body 'Account Approved!' and dashboard link http://localhost:3000/courses. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1]<br>REQ-A-ADM-08: Approval email failure does not undo approval [Version : 1] |
| Keywords: | None |

**Test Case US-77: Approve refuses already approved and admin targets [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Approve refuses already approved and admin targets.

**Test design technique:** Decision Table (target role APPROVED/ADMIN), State Transition invalid

**Requirements:** REQ-A-ADM-07

Preconditions:

Admin API session; original seed Sarah APPROVED and Admin ADMIN; capture their role/approvedAt and email counts.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call admin.approveUser({userId:SarahId}). | BAD_REQUEST 'User is not pending approval'; Sarah unchanged; no approval email. |
| 2 | Call admin.approveUser({userId:AdminId}). | BAD_REQUEST 'User is not pending approval'; Admin remains ADMIN and approvedAt unchanged; no email. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1] |
| Keywords: | None |

**Test Case US-78: Approve rejects nonexistent or empty target ID [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Approve rejects nonexistent or empty target ID.

**Test design technique:** EP (unknown/empty userId), Error Guessing

**Requirements:** REQ-A-ADM-07

Preconditions:

Admin API session; ensure userId 'qa-missing-user' does not exist; capture user counts and mail capture.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call admin.approveUser({userId:'qa-missing-user'}). | NOT_FOUND 'User not found'; no user/count changes and no mail. |
| 2 | Call admin.approveUser({userId:''}). | NOT_FOUND 'User not found'; empty string passes z.string but finds no record; no data changes. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1] |
| Keywords: | None |

**Test Case US-79: Approval remains effective when approval email fails [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Approval remains effective when approval email fails.

**Test design technique:** Error Guessing (SMTP failure), State Transition

**Requirements:** REQ-A-ADM-08, REQ-A-ADM-06, REQ-A-LOG-01

Preconditions:

Admin signed in; Alice PENDING; isolated SMTP stub throws Error('QA SMTP down'); separate anonymous browser ready; restore stub afterward.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call admin.approveUser({userId:AliceId}) while mail fails. | success=true, 'Alice Wong has been approved successfully'; server logs 'Failed to send approval email:'; no rollback. |
| 2 | Inspect Alice then log in as alice@student.utm.my / admin123 in other profile. | Alice APPROVED with approvedAt set; login succeeds and reaches /courses. The document's 'Approved but email failed to send' warning is not implemented and is not claimed as an observed UI message. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1]<br>REQ-A-ADM-08: Approval email failure does not undo approval [Version : 1]<br>REQ-A-LOG-01: Valid credentials route by current role [Version : 1] |
| Keywords: | None |

**Test Case US-80: Sequential replay of approval preserves first approval timestamp [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Sequential replay of approval preserves first approval timestamp.

**Test design technique:** State Transition (PENDING -> APPROVED -> invalid repeat), Error Guessing

| | | |
|---|---|---|
| **Requirements:** REQ-A-ADM-07, REQ-A-ADM-06 | | |
| Preconditions:<br><br>Admin API session; Alice initially PENDING; successful SMTP capture; no concurrent calls; record mail count. | | |
| #: | Step actions: | Expected Results: |
| 1 | Approve Alice once and record approvedAt after success. | 'Alice Wong has been approved successfully'; Alice APPROVED; one approval email attempt. |
| 2 | After first response completes, approve Alice again. | BAD_REQUEST 'User is not pending approval'; first approvedAt unchanged; no extra approval email; Alice not restored to pending list. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-06: Approval confirmation and valid transition [Version : 1]<br>REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1] | |
| Keywords: | None | |

## 1.8.Test Suite : TS-A8 Reject Student Registration

UC-S2-002 rejection: optional reason, final confirmation, deletion, invalid targets, mail failure and re-registration. EP, Decision Table, State Transition, Use Case, Error Guessing.

| | | |
|---|---|---|
| **Test Case US-81: Reject with reason sends explanation and deletes pending applicant [Version : 1]** | | |
| Author: | aland | |
| Summary:<br><br>Reject with reason sends explanation and deletes pending applicant.<br><br>**Test design technique:** State Transition (PENDING -> deleted), Use Case rejection flow<br><br>**Requirements:** REQ-A-ADM-09, REQ-A-ADM-10 | | |
| Preconditions:<br><br>Admin at /admin/approvals; Alice PENDING, captured id and row; SMTP capture succeeds with 2-second delay; no active Alice session. | | |
| #: | Step actions: | Expected Results: |
| 1 | Click Alice's 'Reject'; enter 'Student ID image is unreadable' at 'Enter rejection reason (optional):'. | Then confirmation 'Are you sure you want to reject this user?' appears; no deletion before final confirmation. |
| 2 | Confirm rejection and inspect pending UI/response. | While pending selected action says 'Rejecting...' and both decision buttons disable; success=true, 'Registration for Alice Wong has been rejected'; Alice card removed. |
| 3 | Inspect database and captured rejection mail. | No User with Alice's id/email remains; no REJECTED status record; subject 'UNIShare Registration Update', body 'Registration Status: Not Approved', 'Reason:' and 'Student ID image is unreadable'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1]<br>REQ-A-ADM-10: Rejection email is best effort [Version : 1] | |
| Keywords: | None | |

| Test Case US-82: Optional rejection reason accepts blank and canceled prompt [Version : 1] | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Optional rejection reason accepts blank and canceled prompt.<br><br>**Test design technique:** EP (empty/omitted reason), Decision Table (prompt result x final confirmation)<br><br>**Requirements:** REQ-A-ADM-09, REQ-A-ADM-10 | | |
| Preconditions:<br><br>Admin page; restore independent Alice PENDING fixture before each row; successful SMTP capture; Network panel open. | | |
| #: | Step actions: | Expected Results: |
| 1 | Reject Alice; leave reason empty, choose OK, then confirm final 'Are you sure you want to reject this user?'. | reason='' accepted; 'Registration for Alice Wong has been rejected'; Alice deleted; subject 'UNIShare Registration Update' with no 'Reason:' section. |
| 2 | Restore Alice fixture. Reject again; Cancel the optional reason prompt, then confirm the separate final rejection dialog. | Current UI converts null prompt to omitted reason and still proceeds after final confirmation; same success response and deletion; email omits 'Reason:'. Canceling the reason prompt alone is not final cancellation. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1]<br>REQ-A-ADM-10: Rejection email is best effort [Version : 1] | |
| Keywords: | None | |

| Test Case US-83: Cancel final rejection confirmation preserves applicant [Version : 1] | | |
|---|---|---|
| Author: | aland | |
| Summary:<br><br>Cancel final rejection confirmation preserves applicant.<br><br>**Test design technique:** Decision Table (final confirmation = cancel), State Transition (no transition)<br><br>**Requirements:** REQ-A-ADM-09 | | |
| Preconditions:<br><br>Admin page; Alice PENDING, approvedAt null; user/count snapshots and mail count captured; Network panel open. | | |
| #: | Step actions: | Expected Results: |
| 1 | Click 'Reject'; enter 'QA cancellation check' and accept reason prompt. | Final confirmation says 'Are you sure you want to reject this user?'. |
| 2 | Choose Cancel on final confirmation, then reload admin page. | No admin.rejectUser request; Alice card remains PENDING with same ID and null approvedAt; counts unchanged; no rejection mail. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1] | |
| Keywords: | None | |

| Test Case US-84: Reject refuses approved/admin targets and missing users [Version : 1] | |
|---|---|
| Author: | aland |
| Summary: | |

Reject refuses approved/admin targets and missing users.

**Test design technique:** Decision Table (target APPROVED/ADMIN/missing), State Transition invalid

**Requirements:** REQ-A-ADM-07

Preconditions:

Admin API session; original Sarah APPROVED and Admin ADMIN; userId 'qa-missing-user' absent; capture their rows and mail count.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call admin.rejectUser({userId:SarahId,reason:'QA'}) then repeat with AdminId. | Each returns BAD_REQUEST 'User is not pending approval'; neither user is deleted; roles and approvedAt unchanged; no email. |
| 2 | Call admin.rejectUser({userId:'qa-missing-user'}) then {userId:''} separately. | Each returns NOT_FOUND 'User not found'; no user/count changes and no email. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-07: Missing and nonpending targets rejected [Version : 1] |
| Keywords: | None |

**Test Case US-85: Rejection still deletes applicant when mail fails [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Rejection still deletes applicant when mail fails.

**Test design technique:** Error Guessing (SMTP failure), State Transition (PENDING -> deleted)

**Requirements:** REQ-A-ADM-10, REQ-A-ADM-09

Preconditions:

Admin API session; Alice PENDING; isolated mail stub throws Error('QA SMTP down'); store Alice's id; restore transport afterward.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call admin.rejectUser({userId:AliceId,reason:'Unreadable ID'}) with failing SMTP. | Logs 'Failed to send rejection email:' but returns success=true, 'Registration for Alice Wong has been rejected'. |
| 2 | Inspect database, refresh pending list, then retry reject for old AliceId. | Alice deleted and absent from 'Pending Approvals'; retry returns NOT_FOUND 'User not found'; no retained REJECTED account. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1]<br>REQ-A-ADM-10: Rejection email is best effort [Version : 1] |
| Keywords: | None |

**Test Case US-86: Deleted applicant cannot login and may register again for review [Version : 1]**

| | |
|---|---|
| Author: | aland |

Summary:

Deleted applicant cannot login and may register again for review.

**Test design technique:** State Transition (PENDING -> deleted -> PENDING), Use Case alternative

**Requirements:** REQ-A-ADM-09, REQ-A-LOG-02, REQ-A-REG-16

Preconditions:

Admin API session; Alice initially PENDING; fresh anonymous browser has never logged in as Alice; controlled SMTP. Record old AliceId. Use new registration password Test1234, not seed admin123.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | As Admin reject Alice and confirm; in fresh anonymous profile attempt alice@student.utm.my / admin123 login. | Rejection response 'Registration for Alice Wong has been rejected'; login shows 'Invalid email or password'; no session established. |
| 2 | On /signup register name Alice Wong, same alice@student.utm.my email, Test1234/confirmation, seeded UTM/Computing and newly uploaded valid ID. | 'Registration Successful!'; exactly one new Alice user with a different id, role PENDING and approvedAt null; email uniqueness does not prevent reapplication after deletion. |
| 3 | Log in with alice@student.utm.my / Test1234. | Ends at /waiting-approval with 'Account Pending Approval'; prior rejection does not bypass new admin review. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-A-REG-16: Persist pending applicant with bcrypt and success UI [Version : 1]<br>REQ-A-ADM-09: Rejection confirmation, optional reason and deletion [Version : 1]<br>REQ-A-LOG-02: Generic invalid-credentials errors [Version : 1] |
| Keywords: | None |